6.1 Introduction – Easy Explanation

Probability distributions are needed in simulation because when we simulate real-world systems like a bank, factory, or shipping, some things are random. Examples of random things include time between arrivals (e.g., customers at a bank), processing times (e.g., machines in a factory), and demand sizes (e.g., how many items people buy). To simulate these random events correctly, we must describe them using probability distributions. For example, in a single-server queue (one machine serving customers), interarrival times can follow an exponential distribution with a mean of 1 minute. In inventory simulation, demand sizes can be 1, 2, 3, or 4 items, with specific probabilities like 1/6, 3/6, etc.

Once we know the distributions, we can generate random numbers from these distributions to simulate the system over time. Later chapters explain how to generate random numbers (Chapters 7 and 8). This chapter focuses on how to choose the correct probability distributions for the inputs.

Randomness exists in all real systems. Real systems always have randomness. Examples include:

- **Manufacturing** → processing times, machine failure, repair times
- **Defense** → arrival of missiles, outcomes of battles
- **Communications** → time between messages, message size/type
- **Transportation** → ship-loading times, arrival times of shipments

Histograms (graphs showing data frequency) from real projects often don't look like a normal distribution. Positive skew has a long tail on the right, and negative skew has a long tail on the left.

Choosing the correct distribution matters because just knowing the average (mean) is not enough; you need the full distribution. Using the wrong distribution can give wrong simulation results.

For example, in a single-server queue system, one machine serves customers who arrive randomly with exponential interarrival times. Service times of 200 customers are collected, but the underlying distribution is unknown. Five possible distributions were tested:

- Exponential
- Gamma
- Weibull
- Lognormal
- Normal

100 simulation runs were done for each distribution, collecting 1000 delays in the queue per run.

The simulation results can be summarized as follows:

- **Exponential** → Avg. delay in queue = 6.71 min, Avg. number in queue = 6.78, Proportion of delays ≥ 20 = 0.064

- **Gamma** → Avg. delay in queue = 4.54 min, Avg. number in queue = 4.60, Proportion of delays ≥ 20 = 0.019
- **Weibull** → Avg. delay in queue = 4.36 min, Avg. number in queue = 4.41, Proportion of delays ≥ 20 = 0.013
- **Lognormal** → Avg. delay in queue = 7.19 min, Avg. number in queue = 7.30, Proportion of delays ≥ 20 = 0.078
- **Normal** → Avg. delay in queue = 6.04 min, Avg. number in queue = 6.13, Proportion of delays ≥ 20 = 0.045

The best match is the **Weibull distribution**, as its average delay is closest to reality (4.36 min). Using the wrong distribution, like Lognormal, can overestimate delays (7.19 min), which is a 65% error. Even distributions that look similar can give very different results because of differences in the tails (extreme values).Key takeaways for exams:

- Always try to fit the best distribution for random inputs.
- Wrong choice results in wrong simulation output → wrong decisions.
- The mean alone is not enough; the full probability distribution matters.

Easy memory tip: "*Random inputs need the right distribution, or simulation results can be way off.*"

Specifying Distributions from Data – Easy Explanation

When you have data for a random input like service times or arrival times, you can use the data in three main ways, from least ideal to most ideal.

1. Use data values directly (Trace-driven simulation)

In this method, you just take the actual data points and use them in the simulation whenever needed. For example, if a customer service time is 5 minutes, you use 5 minutes in the simulation.

Drawbacks:

- You can only simulate what already happened (no new situations).
- Usually, there is not enough data for long simulations.

When useful:

- Comparing a new system with an existing system using historical data.
- Validating the model (checking if simulation matches real system).

2. Use an empirical distribution

Here, you create a distribution from the data itself. Instead of using single values, you sample from this data-based distribution. This method can generate any value between the minimum and maximum of the data, making it more flexible than just replaying old data. Empirical distributions are usually better than method 1, especially for continuous data.

3. Fit a theoretical distribution (Preferred)

This method uses statistics to find a theoretical distribution that fits your data. Examples include Exponential, Poisson, Normal, Weibull, and Lognormal. You can check if the chosen distribution fits well using hypothesis tests.

Advantages:

- Smooths out irregularities in the data (good for small datasets).
- Can generate values outside the observed range, which is important for extreme events.
- Easier to adjust parameters like mean or rate.
- Compact – no need to store every data point.
- Sometimes physical reasons exist for choosing a particular distribution.

Drawbacks:

- Not always possible if data is a mix of different populations.
- Difficult if data is heavily rounded or discrete with few values.
- Theoretical distributions can sometimes generate unrealistically large values, so truncation may be needed.

Summary for Exam

- If no theoretical fit → use empirical distribution.
- If theoretical fit exists → use theoretical distribution (more accurate, flexible, compact).
- Trace-driven (raw data) → only for validation or comparing old vs new system.

6.2.1 Parameterization of Continuous Distributions

A distribution such as Normal or Gamma can be described using parameters. These parameters tell us how the distribution looks and behaves.

Three types of parameters

- **Location (g)** – shifts the distribution left or right on the x-axis
  Example: Mean (m) of Normal distribution
- **Scale (b)** – stretches or compresses the distribution
  Example: Standard deviation (σ) of Normal distribution

- **Shape (a)** – changes the basic shape of the distribution
  Example: Skewness in Beta distribution

The location parameter moves the distribution without changing its shape. For example, if a variable X has location 0, then Y = X + g has location g.

The scale parameter changes the width or spread without changing the shape. For example, if X has scale 1, then Y = bX has scale b.

The shape parameter changes the form of the curve itself. Some distributions like Normal or Exponential do not have shape parameters, but distributions like Beta or Gamma do.

6.2.2 Continuous Distributions – Easy English

There are many continuous distributions used in simulation.

Table 6.3 in the book lists 13 distributions along with the following information:

- Applications – what the distribution is used for
- Density function – formula for probabilities
- Parameters – location, scale, shape
- Range – possible values
- Mean, variance, mode – summary statistics
- MLE – maximum likelihood estimator used to estimate parameters from data
- Comments – relationships with other distributions

Some less familiar distributions are included because they often fit data better than standard ones, such as Beta, Johnson SB, Johnson SU, Log-logistic, Pearson V, and Pearson VI.

Example: Uniform Distribution U(a, b)

The uniform distribution is used when we know a variable can take any value between a and b, but we do not know much else about how values are distributed. The special case U(0,1) is very important for generating random numbers for other distributions.

Density function (f(x))

The density function is $f(x) = 1 / (b - a)$ for $a \le x \le b$, and $f(x) = 0$ otherwise. The graph is a flat line because all values between a and b are equally likely.

Distribution function (F(x))

The distribution function gives the probability that $X \le x$.
$F(x) = 0$ if $x < a$,
$F(x) = (x - a) / (b - a)$ if $a \le x \le b$,
$F(x) = 1$ if $x > b$.

Parameters

- a = location (start of range)
- b – a = scale (width of range)

Range-[a, b]

Mean-(a + b) / 2

Variance-$(b – a)^2$ / 12

Mode-There is no mode because all values are equally likely.

MLE (estimators)

- a = minimum of data values
- b = maximum of data values

U(0,1) is a special case of the Beta distribution. The graph is a flat line between a and b because the density is constant.

"Location moves it, scale stretches it, shape changes its form. Uniform is flat, U(a,b), all values between a and b equally likely."

1. Exponential Distribution – expo(b)

- Time between arrivals of customers at a system (bank, call center, etc.)
- Time to failure of a machine or equipment

Property	Value / Explanation
Density f(x)	f(x) = 1/b * e^(-x/b), x ≥ 0
Distribution F(x)	F(x) = 1 – e^(-x/b), x ≥ 0
Parameter	Scale b > 0
Range	[0, ∞)
Mean	b
Variance	b^2
Mode	0 (most frequent value)
MLE (estimator)	Mean of the sample: $\hat{b} = \Sigma X_i / n$

Special points:

- A special case of **Gamma** and **Weibull** (shape a = 1)
- Only continuous distribution with **memoryless property** (past doesn't affect future)

- Sum of m independent expo(b) → **Gamma / Erlang distribution**

2. Gamma Distribution – gamma(a, b)

- Time to complete a task: customer service, machine repair, etc.

Property	Value / Explanation
Density f(x)	$f(x) = (1/(b^a * \Gamma(a))) * x^{(a-1)} * e^{(-x/b)}$, $x > 0$
Distribution F(x)	No simple formula unless a is integer: sum formula used
Parameters	Shape $a > 0$, Scale $b > 0$
Range	$[0, \infty)$
Mean	$a * b$
Variance	$a * b^2$
Mode	$b * (a - 1)$, if $a \geq 1$; 0 if $a < 1$

Special points:

- expo(b) = gamma(1, b)
- gamma(m, b) = m-Erlang(b) if m is integer
- Chi-square distribution = gamma(k/2, 2)
- Adding independent gamma variables → new gamma with sum of shapes

3. Weibull Distribution – Weibull(a, b)

- Time to complete a task or equipment failure
- Often used **if no data is available** (rough model)

Property	Value / Explanation
Density f(x)	$f(x) = (a/b) * (x/b)^{(a-1)} * e^{(-(x/b)^a)}$, $x > 0$
Distribution F(x)	$F(x) = 1 - e^{(-(x/b)^a)}$, $x > 0$
Parameters	Shape $a > 0$, Scale $b > 0$
Range	$[0, \infty)$
Mean	$b * \Gamma(1 + 1/a)$
Variance	$b^2 * [\Gamma(1 + 2/a) - (\Gamma(1 + 1/a))^2]$
Mode	$b * ((a - 1)/a)^{(1/a)}$ if $a \geq 1$; 0 if $a < 1$

Special points:

- expo(b) = Weibull(1, b)

- log(Weibull) → **Gumbel (Extreme Value) distribution**
- Weibull(2, b) = **Rayleigh distribution**
- Large a → sharp peak at mode
- Negative skew when $a > 3.6$

Easy memory tips for exams:

- **Exponential:** "Memoryless, simplest, time between arrivals, mean = b"
- **Gamma:** "Sum of exponentials, shape matters, mean = a*b"
- **Weibull:** "Flexible shape, can model many types of failure or task times"

1. Normal Distribution – N(m, s^2)

- Errors (e.g., impact points, measurement errors)
- Sum of many small effects → Central Limit Theorem

Property	Value / Explanation
Density f(x)	$f(x) = (1 / \sqrt{(2\pi s^2)}) * e^{\wedge}(-(x-m)^2 / 2s^2)$, $x \in (-\infty, \infty)$
Parameters	Location m (mean), Scale $s > 0$ (standard deviation)
Range	$(-\infty, \infty)$
Mean	m
Variance	s^2
Mode	m
MLE	$\hat{m}$ = mean of sample, $\hat{s}^2$ = sample variance

- Standard normal: N(0, 1)
- Linear combinations of normals → also normal
- Independent standard normals → sum of squares = chi-square distribution
- e^X → lognormal distribution
- If used for nonnegative quantities, may need truncation at $x = 0$

2. Lognormal Distribution – LN(m, s^2)

Use / Applications:

- Product of many positive quantities (Central Limit for logs)

- Time to complete a task (similar to gamma or Weibull)
- Often used as a rough model

Property	Value / Explanation
Density f(x)	$f(x) = (1 / (x\sqrt{(2\pi s^2)})) * e^{(-(\ln x - m)^2 / 2s^2)}$, $x > 0$
Parameters	Shape $s > 0$, Scale $e^m > 0$
Range	$[0, \infty)$
Mean	$e^{(m + s^2/2)}$
Variance	$(e^{(s^2)} - 1) * e^{(2m + s^2)}$
Mode	$e^{(m - s^2)}$
MLE	ln(data) treated as normal for estimating $\hat{m}$, $\hat{s}$

Special points:

- $\ln(X) \sim \text{Normal}(m, s^2)$ → useful for fitting data
- As $s \to 0$, becomes degenerate at e^m (sharp peak)

3. Beta Distribution – beta(a1, a2)

- Random proportion (e.g., defect rate)
- Time to complete a task in PERT
- Rough model when no data available

Property	Value / Explanation
Density f(x)	$f(x) = (1 / B(a1, a2)) * x^{(a1-1)} * (1-x)^{(a2-1)}$, $0 < x < 1$
Parameters	Shape $a1 > 0$, $a2 > 0$
Range	$[0, 1]$
Mean	$a1 / (a1 + a2)$
Variance	$a1\ a2 / [(a1 + a2)^2 (a1 + a2 + 1)]$
Mode	Depends on a1, a2: see rules below
MLE	Numerical solution using digamma function

Mode rules:

- $a1 > 1, a2 > 1$ → mode exists: $(a1-1)/(a1+a2-2)$
- $a1 < 1, a2 < 1$ → mode at 0 and 1
- $a1 = a2$ → symmetric about 0.5

Special points:

- Beta(1, 1) = Uniform(0, 1)
- Can rescale [0, 1] → [a, b]
- Ratio of gammas → beta distribution

6.2.3 Discrete Distributions

Discrete distributions deal with countable values. The values are usually integers like 0, 1, 2, 3, and so on. Examples include number of students, number of defective items, and number of successes. In Table 6.4, the book explains six discrete distributions in the same format as continuous distributions, including where they are used, the formula, mean, variance, and comments.

1. Bernoulli Distribution – Bernoulli(p)

The Bernoulli distribution represents a single experiment with only two possible outcomes: success (1) or failure (0). Examples include tossing a coin (head or tail), a student passing or failing, or a machine working or failing.

- Values: 0 or 1 only
- Parameter: p = probability of success ($0 < p < 1$)
- Mean: p
- Variance: $p(1 - p)$

Exam tip: Bernoulli means a single yes/no experiment.

2. Discrete Uniform Distribution – DU(i, j)

The discrete uniform distribution is used when several integer values are possible and all values are equally likely. Examples include a random number between 1 and 6 when rolling a dice, or any integer between i and j.

- Values: $i, i+1, i+2, \dots, j$
- Mean: $(i + j) / 2$
- Variance: $((j - i + 1)^2 - 1) / 12$

Exam tip: If all outcomes have the same probability, it is a discrete uniform distribution.

3. Binomial Distribution – bin(t, p)

The binomial distribution represents repeated Bernoulli trials and counts the number of successes. Examples include number of heads in 10 coin tosses, number of defective items in a batch, or number of students who pass out of 50.

- Parameters:

- t = number of trials
- p = probability of success

- Values: $0, 1, 2, \dots, t$
- Mean: tp
- Variance: $tp(1 - p)$

Exam tip: Binomial means many yes/no experiments.Relationship (Very Important for Exam): Bernoulli is a special case of Binomial. Bernoulli(p) = bin(1, p).

6.2.4 Empirical Distributions (Very Easy)

An empirical distribution is created directly from real data. No theoretical formula is used. It is used when a theoretical distribution does not fit the data well or when fitting is too difficult. Sometimes data is irregular and no standard distribution matches it properly. Instead of forcing a formula, we use the data itself.

Two Data Situations

Case 1: Original data available

You have data values $X_1, X_2, X_3, \dots, X_n$.

- Steps:
  - Sort the data from smallest to largest
  - Build a piecewise-linear distribution function $F(x)$
  - Generate random values between data points

This creates a smooth curve based on actual data.

Case 2: Grouped data (Histogram data)

In this case, individual values are not available. Only the number of values within intervals is known. For example, 0–5 may contain 10 values and 5–10 may contain 20 values. This is called grouped data. The distribution is built using interval frequencies.

Empirical vs Theoretical

- Empirical Distribution:
  - Uses real data
  - No assumptions
  - Good when theoretical fit fails

- Theoretical Distribution:
  - Uses formulas
  - Assumes known shape
  - Preferred when fit is good

If theory fails, use empirical.
If theory fits, use theoretical.

One-Line Exam Answers

- Discrete distribution: Values are countable integers
- Bernoulli: One trial, two outcomes
- Binomial: Many Bernoulli trials
- Empirical distribution: Built directly from observed data

Poisson Distribution – Poisson(λ)

Poisson distribution is used to count how many times something happens in a fixed time or space when events happen randomly, at a constant average rate, and independently. It is mainly used when we are counting the number of events in a fixed interval.

Poisson distribution is commonly used for counting events such as the number of customers arriving in one hour, the number of phone calls in ten minutes, the number of defects in a product, or the number of items demanded from inventory. The main exam keyword is "number of events in a fixed interval".

Conditions for Poisson Distribution

- Events happen randomly
- Events happen at a constant average rate
- Events are independent

Values (Range):X = 0, 1, 2, 3, …

Parameter

- λ (lambda) = average number of events
- $\lambda > 0$

If the average arrivals per hour are 5, then $\lambda = 5$.

Probability Mass Function

- $p(x) = (e^{-\lambda} \lambda^x) / x!$

You do not need to derive this formula in exams, but you should recognize it.

Mean and Variance

- Mean = λ
- Variance = λ

In Poisson distribution, the mean and variance are equal. This is a very important exam point.

Mode

- If λ is an integer → λ and $\lambda - 1$
- Otherwise → floor value of λ

MLE (Maximum Likelihood Estimator)

- Estimated λ = sample mean
- $\hat{\lambda} = \bar{X}$

1. Additivity Property

- If $X_1 \sim \text{Poisson}(\lambda_1)$
- If $X_2 \sim \text{Poisson}(\lambda_2)$
- If X_1 and X_2 are independent
- Then $X_1 + X_2 \sim \text{Poisson}(\lambda_1 + \lambda_2)$

The sum of independent Poisson variables is also Poisson.

2. Relation with Exponential Distribution

If interarrival times are exponential, then the number of arrivals follows a Poisson distribution. This is why Poisson distribution is widely used in queueing systems.

Empirical Distribution

An empirical distribution is constructed directly from observed data instead of using a theoretical formula. It is used when a theoretical distribution does not fit well or when fitting is too difficult. In empirical distributions, the shape depends entirely on the given data.

Case 1: Original (Ungrouped) Continuous Data Available

When original data values $X_1, X_2, \ldots, X_n$ are available, we construct the distribution using the data itself. The distribution rises faster where data is dense and rises slowly where data is sparse.

- Sort data in increasing order: X(1) ≤ X(2) ≤ … ≤ X(n)
- Build a piecewise-linear distribution function
- Connect points with straight lines

The generated values will always be greater than or equal to the smallest observed value and less than or equal to the largest observed value. The mean of this empirical distribution is not equal to the sample mean, which is a disadvantage. Another disadvantage is that values outside the observed range cannot be generated.

Case 2: Grouped Continuous Data

When individual data values are not available and only interval information is given, such as [0, 5) containing 10 values and [5, 10) containing 20 values, the empirical distribution is constructed using interval frequencies. The distribution rises faster where more data exists and generated values are bounded between the smallest and largest interval boundaries.

- Use interval boundaries $a_0, a_1, \ldots, a_k$
- Compute cumulative proportions
- Join points using straight lines

Problem with Empirical Distributions

In real-life situations, many distributions are right-skewed, meaning large values appear in the right tail. Large values may not appear in small samples, but they can strongly affect simulation results, such as long service times causing long queues.

To handle this issue, an exponential tail can be attached to the right side of the empirical distribution. This allows values larger than the maximum observed value and produces more realistic simulation results.

Empirical Distribution for Discrete Data

When original discrete data is available, the probability for each value is calculated directly from its frequency in the sample.

If Original Discrete Data is Available

- For each value x:
  - p(x) = (number of times x appears) / n

If Discrete Data is Grouped

- Only total probability in each interval is known
- Distribution inside interval is chosen arbitrarily

The allocation of probability inside an interval is not unique.

Poisson

- Counts events in fixed time or space
- Parameter λ
- Mean = Variance = λ

Empirical Distribution

- Built directly from data
- Used when theoretical distribution does not fit
- Piecewise-linear
- Bounded unless exponential tail is added

6.3 Techniques for Assessing Sample Independence

What does "sample independence" mean?

The observations $X_1, X_2, \ldots, X_n$ are independent if knowing one value does not affect the others. In other words, one observation gives no information about another observation.

For example, tossing a coin many times produces independent results because each toss does not affect the next one. However, hourly temperature measurements are not independent because nearby hours usually have similar temperatures. Similarly, waiting times of customers in a busy queue are not independent because congestion can cause consecutive delays to be related.

Why is independence important?

Many statistical methods assume independence. If the data are not independent, these methods may produce incorrect results. Methods such as Maximum Likelihood Estimation (MLE) and chi-square tests require independence of observations. However, simple graphical tools like histograms can still be used even when the data are dependent.

Why data collected over time may be dependent

When data are collected over time, they are often related to nearby values. For example, hourly temperature readings are positively correlated because temperatures change gradually. In a queueing system, if the system becomes congested, one long delay may cause the next delay to also be long, which creates positive correlation.

Graphical Methods to Check Independence

These methods are informal and visual. They are important for understanding dependence patterns in exams.

1. Correlation Plot

A correlation plot shows the estimated sample correlations $\hat{r}_j$ for different lags $j = 1, 2, \ldots, l$. The lag j represents how far apart two observations are in time. For example, $\hat{r}_1$ measures the correlation between X_1 and X_2, while $\hat{r}_2$ measures the correlation between X_1 and X_3.

If the data are independent, the true correlation should be zero. In practice, estimated correlations will not be exactly zero, but they should be close to zero.

Interpretation Rules

- Small correlation values near 0 → independence likely
- Large positive or negative values → dependence likely

2. Scatter Diagram

A scatter diagram plots pairs of consecutive observations (X_i, X_{i+1}). The pattern of points helps identify dependence.

If the points form a random cloud, the data are likely independent. If the points form an upward-sloping pattern, there is positive correlation. If the points form a downward-sloping pattern, there is negative correlation.Pattern Interpretation

- Random cloud → Independent
- Upward line → Positive correlation
- Downward line → Negative correlation

Example 6.2 (Independent Data)

In this example, data are generated from an exponential distribution. The correlation values are approximately zero, and the scatter plot appears random. This confirms independence.

Example 6.3 (Dependent Data)

In this example, queueing delays are observed with traffic intensity $\rho = 0.8$. The correlation values are high for small lags, and the scatter plot shows an upward trend. This confirms positive dependence.

Formal (Statistical) Tests for Independence

These are statistical tests used to formally test independence.

1. Rank von Neumann Test

- Proposed by Bartels
- Nonparametric (no distribution assumption)
- Assumes no ties (no equal values)
- Problematic for discrete data

The rank von Neumann test is generally more powerful than the runs test, but it does not work well when there are ties in the data.

2. Runs Test

- Nonparametric
- Works better when ties exist
- Less powerful than rank von Neumann test in some cases

The runs test is useful when the data contain equal values, especially in discrete distributions.

6.4 Activity I: Hypothesizing Families of Distributions

Before estimating parameters, we first decide which distribution family might fit the data. Examples of distribution families include Normal, Exponential, and Poisson distributions.

Sometimes distributions are chosen based on theoretical knowledge even without data. For example, interarrival times are often modeled using the exponential distribution, and the number of arrivals is often modeled using the Poisson distribution. Service time should not be modeled using a Normal distribution because the Normal distribution allows negative values. Proportions are often modeled using the Beta distribution because values must lie between 0 and 1.

When theory alone is not enough, heuristics based on the data are used. These include examining summary statistics, histograms, and the overall shape of the data such as whether the distribution is symmetric or skewed.

6.4.1 Summary Statistics

Summary statistics are numerical values calculated from data that help identify the appropriate distribution.

If the mean is approximately equal to the median, the distribution is likely symmetric. The Normal distribution is a common example of a symmetric distribution.

2. Coefficient of Variation (CV) for Continuous Data

The coefficient of variation is defined as the standard deviation divided by the mean.

Interpretation

- CV ≈ 1 → Exponential distribution
- CV < 1 → Gamma or Weibull distribution with shape parameter greater than 1
- CV > 1 → Lognormal distribution (often)
- CV not defined if mean = 0

3. Lexis Ratio (τ) for Discrete Data

The Lexis ratio is defined as variance divided by mean.

Interpretation

- $\tau = 1$ → Poisson distribution
- $\tau < 1$ → Binomial distribution
- $\tau > 1$ → Negative binomial or Geometric distribution

This is very important for exams.

4. Skewness

Skewness measures the symmetry of the distribution.

- Skewness = 0 → Symmetric
- Skewness > 0 → Right-skewed
- Skewness < 0 → Left-skewed

Most real-life data are right-skewed.

5. Kurtosis

Kurtosis measures tail heaviness of a distribution. However, kurtosis is generally not very helpful in choosing distributions.

Independence

- Required for MLE and chi-square tests
- Checked using correlation plot and scatter diagram

Distribution Selection

- Use theory and data
- Check range, shape, and skewness

Key Identification Rules

- CV ≈ 1 → Exponential
- Lexis ratio = 1 → Poisson
- Lexis ratio < 1 → Binomial
- Lexis ratio > 1 → Negative binomial

6.4.2 Histograms

A histogram is a graph used to understand the shape of data. It helps us guess which probability distribution (such as Normal or Exponential) might fit the data. For continuous data, a histogram gives a rough visual picture of the density curve. In exams, remember that a histogram is mainly a visual tool used to identify the distribution family.

Why Histograms Are Useful

Many probability distributions have recognizable shapes, such as bell-shaped or right-skewed. By looking at the shape of the histogram, we can make an initial guess about the correct distribution. At this stage, we focus only on the overall shape of the data, not on exact numerical values like mean or variance.

How to Construct a Histogram

- Take the full data range (smallest value to largest value)
- Divide the range into k equal intervals (bins)
- Ensure all bins have the same width
- Count how many data points fall in each bin
- Draw bars with data values on the x-axis and proportions (or frequencies) on the y-axis

The result is a bar graph called the histogram.

Meaning of the Histogram Function

The histogram function is zero outside the data range and constant within each interval. Because of this, the graph looks step-shaped. A true density curve is smooth, while a histogram is piecewise constant. However, the histogram still approximates the shape of the density.

Why Histogram Shape Matches Density Shape

The proportion of data inside an interval is approximately equal to the probability of that interval. Probability corresponds to the area under the density curve. Therefore, after proper scaling, the height of each histogram bar roughly matches the height of the density curve in that region. As a result, both graphs have similar overall shapes.

Choosing the Number of Intervals (k)

Choosing the number of intervals is the main difficulty when constructing histograms.

- If intervals are too small → the histogram looks rough and noisy
- If intervals are too large → the histogram looks blocky and hides important details

The best practical approach is to try different interval widths and choose the smallest width that still produces a smooth and meaningful shape.

Rules of Thumb

- Sturges' rule: $k = 1 + \log_2 n$
- This rule is simple but not always reliable
- Practical judgment and experimentation are usually better

Histogram for Discrete Data

For discrete data, each possible value has its own bar, and the height of the bar equals the proportion of times that value appears. This directly estimates the probability mass function (PMF). Unlike continuous data, no interval width decision is required, so the process is simpler.

Multiple Peaks (Multimodal Histograms)

Sometimes a histogram shows more than one hump. This usually means the data come from different sources or different situations.

For example, machine repair times may include short times for small repairs and long times for major repairs. This creates two peaks. In such cases, the data can be divided into groups, and a separate distribution can be fitted to each group. These distributions are then combined into a mixture distribution.

- Histogram helps guess distribution family
- Shape is more important than exact numerical values
- Interval width choice is subjective
- Too small width → noisy graph
- Too large width → oversmoothing
- Works for both continuous and discrete data

- Multiple humps suggest mixture distributions

6.4.3 Quantile Summaries and Box Plots

A quantile summary is a short numerical summary of data. It helps us understand whether the data are symmetric, skewed to the right, or skewed to the left. It can be used for both continuous and discrete data, although here it is explained using continuous data. In exams, remember that a quantile summary is mainly used to detect symmetry or skewness.

Important Quantiles

- Median ($x_{0.5}$) → middle value
- Lower quartile ($x_{0.25}$) → 25% of data below
- Upper quartile ($x_{0.75}$) → 75% of data below
- Lower octile ($x_{0.125}$) → 12.5% of data below
- Upper octile ($x_{0.875}$) → 87.5% of data below
- Extremes → minimum and maximum

A quantile divides data into specific proportions.

Structure of a Quantile Summary

A quantile summary includes the median, quartiles, octiles, and the extreme values (minimum and maximum). For each level, matching values from the left and right sides are taken, and their midpoint is calculated. These midpoints help determine the shape of the distribution.

How Quantile Summary Detects Shape

Case 1: Symmetric Distribution

If the distribution is symmetric, all midpoints are approximately equal. A normal distribution is a common example of a symmetric distribution.

Case 2: Right-Skewed Distribution

If the distribution is right-skewed, the midpoints increase from top to bottom. This means the right tail is longer. The exponential distribution is a common example of a right-skewed distribution.

Case 3: Left-Skewed Distribution

If the distribution is left-skewed, the midpoints decrease from top to bottom. This means the left tail is longer.

This midpoint pattern is very important for exams.

What is a Box Plot?

A box plot is a graphical version of the quantile summary. It displays the median, quartiles, and the overall spread of the data. The box represents the middle 50% of the data, from the lower quartile ($x_{0.25}$) to the upper quartile ($x_{0.75}$). Some box plots may not show octiles, depending on the presentation.

How to Read a Box Plot

- Longer right side → right-skewed distribution
- Longer left side → left-skewed distribution
- Box centered evenly → symmetric distribution

Example 6.4: Interarrival Time Data

In this example, car interarrival times at a bank were recorded, with 219 observations. From theoretical reasoning, cars arrive one by one, the arrival rate is approximately constant, and arrivals are independent. Therefore, an exponential distribution is expected.

The summary statistics support this conclusion.

- Mean greater than median → indicates right skew
- Skewness = 1.478 → strong right skew
- CV approximately equal to 1 → consistent with exponential distribution

The histogram was drawn using different bin widths. The smoothest histogram resembled the exponential shape. Sturges' rule produced bins that were too wide.

The quantile summary showed that midpoints increased downward, and the box plot showed a longer right side. All evidence confirmed that the interarrival times follow an exponential distribution.

The final conclusion is that theory, summary statistics, histogram, and quantile summary together indicate an exponential distribution.

Example 6.5: Discrete Demand Data

In this example, weekly demand values were observed. The data were discrete and included many zero demands.

The summary statistics indicated the following:

- Lexis ratio greater than 1 → not Poisson or Binomial
- Skewness greater than 0 → right-skewed
- Histogram decreasing → many small values

Possible distributions included geometric or negative binomial. Since the histogram showed a monotonically decreasing shape, the best match was the geometric distribution.

- Quantile summary detects skewness
- Box plot is graphical quantile summary
- Increasing midpoints → right skew
- Decreasing midpoints → left skew
- Nearly equal midpoints → symmetric distribution
- Always combine theory, summary statistics, histogram, and box plot when selecting a distribution

ACTIVITY II: ESTIMATION OF PARAMETERS

In Activity I, we only guessed the type of distribution, such as exponential or normal. However, a distribution is not complete unless we know its parameter value, such as the mean, rate, or variance. Finding the numerical value of an unknown parameter using data is called parameter estimation.

What is an Estimator?

An estimator is a formula based on sample data. It gives us an estimate of an unknown parameter. For example, if the parameter is the population mean, the estimator is the sample mean.

Why Maximum Likelihood Estimation (MLE)?

There are many estimation methods, but we focus on MLE for the following reasons:

- MLE usually gives good and reliable estimates
- MLE is needed later for the chi-square test
- MLE has a very logical idea: choose the parameter value that makes the observed data most likely

The basic idea of maximum likelihood is that the best parameter is the one that best explains the data we observed.

Discrete Case (Concept Only)

Suppose we have sample data $(X_1, X_2, \ldots, X_n)$, and the distribution has one unknown parameter (u) with probability function $p_u(x)$. The likelihood function gives the probability of obtaining the observed data for a given value of (u).

The likelihood function is:
L(u) = p_u(X_1), p_u(X_2), …, p_u(X_n)

The maximum likelihood estimator (MLE) is defined as the value of (u) that maximizes L(u).

Continuous Case (Same Idea)

In the continuous case, we use the probability density function $f_u(x)$. The likelihood function becomes:
L(u) = f_u(X_1), f_u(X_2), …, f_u(X_n)

The MLE is the value of (u) that maximizes L(u).

Why Use Log-Likelihood?

The likelihood function involves products, which are difficult to differentiate. Taking the logarithm converts products into sums, making differentiation easier. Since the logarithm is an increasing function, maximizing the log-likelihood is equivalent to maximizing the likelihood.

- Parameter estimation means finding the parameter value from data
- MLE is the value that maximizes the likelihood
- Likelihood is the joint PDF or PMF of the observed data
- Log-likelihood is used to simplify calculation
- For exponential distribution, the MLE of (b) equals the sample mean

6.6 ACTIVITY III

Determining How Representative the Fitted Distributions Are

Activity III focuses on checking how well the fitted distribution represents the real data. In Activity I, we guessed the type of distribution. In Activity II, we estimated the parameter values. Now in Activity III, we verify whether the chosen distribution is good enough to describe the data. The main goal is not to find a perfect distribution, because no fitted distribution is ever perfect. Instead, we want a distribution that is accurate enough for simulation purposes.

Two ways to check goodness of fit

- Heuristic (graphical) methods
- Goodness-of-fit tests

In this section, we focus only on heuristic or graphical methods.

6.6.1 Heuristic (Graphical) Procedures

There are five graphical methods, but only the main ones are explained here.

Density–Histogram Plot (Continuous Data)

In this method, we draw a histogram of the observed data and then draw the fitted probability density function on the same graph. After that, we visually compare their shapes. If the histogram bars and the fitted density curve look similar, then the fitted distribution is considered a good fit.

In the frequency comparison for the continuous case, the data are divided into k intervals. For each interval, h_j represents the observed proportion of data, and r_j represents the expected proportion from the fitted distribution. The expected proportion is calculated using the integral
$r_j = \int_{b_{j-1}}^{b_j} \hat{f}(x)\,dx$.
If h_j and r_j are approximately equal, then the fit is considered good.

For discrete data, h_j represents the observed proportion at value x_j, and r_j represents the fitted probability $\hat{p}(x_j)$. If the bars for observed and expected values almost match, then the fit is good.

If the fitted distribution is good and the sample size is large, then $h_j \approx r_j$.

In Example 6.9, the data were interarrival times and the fitted distribution was exponential with parameter $\hat{b} = 0.399$. The histogram and the density curve matched well, so it was considered a good fit.

In Example 6.10, the data were demand sizes and the fitted distribution was geometric with $\hat{p} = 0.346$. The bars matched except for a small difference at $x = 1$, but it was still considered acceptable.

Distribution-Function-Differences Plot

This plot compares cumulative probabilities instead of individual probabilities. The empirical distribution function is defined as
$F_n(x) = (\text{Number of } X_i \le x) / n$.
This means it represents the proportion of data less than or equal to x.

In this method, we plot $\hat{F}(x) - F_n(x)$. If the fitted distribution is perfect, the plot would be a horizontal line at zero. Larger deviations from zero indicate a worse fit. Often, problems appear at the lower end of the data. Smaller deviations from zero mean a better fit.

In Examples 6.11 and 6.12 for the exponential and geometric cases, the deviations were mostly small, so the fit was considered good overall.

Probability Plots

Instead of comparing curves, probability plots check whether the points form a straight line. Humans are generally better at detecting straight-line patterns than curve differences.

(a) Q–Q Plot (Quantile–Quantile)

In a Q–Q plot, we compare observed quantiles with model quantiles. First, we sort the data as $X_{(1)}, X_{(2)}, \ldots, X_{(n)}$. Then we compute $q_i = (i - 0.5) / n$. After that, we plot the observed quantiles against the model quantiles. If the points form a straight line with slope approximately equal to 1 and intercept approximately equal to 0, then the fit is good. The Q–Q plot is especially useful for detecting differences in the tails of the distribution.

In Example 6.13 for the exponential distribution, the Q–Q plot showed slight deviation at the high end, but overall the middle fit very well.

(b) P–P Plot (Probability–Probability)

In a P–P plot, we compare model cumulative distribution function values with empirical cumulative distribution function values. We plot $(\tilde{F}n(X(i)), \hat{F}(X(i)))$. If the points form a straight line, then the fit is good. The P–P plot is more sensitive to differences in the middle of the distribution.

In Example 6.14 for the geometric distribution, the P–P plot was reasonably straight, indicating good agreement.

Q–Q vs P–P

- Q–Q plot → Best at detecting tail differences
- P–P plot → Best at detecting middle differences

FINAL SUMMARY

Activity III checks how good the fitted distribution is. A perfect fit is impossible, so we look for a distribution that is good enough. Heuristic methods are visual checks used to compare the fitted distribution with observed data.

The main graphical tools are:

- Density–histogram plot
- Frequency comparison
- Distribution-function-difference plot
- Q–Q plot
- P–P plot

If the Q–Q or P–P plot forms a straight line, the fit is considered good. Q–Q plots focus more on tail behavior, while P–P plots focus more on the middle of the distribution.

6.6.2 Goodness-of-Fit Tests

A goodness-of-fit test is a formal statistical test used to check whether the observed data really come from the fitted distribution or not. It is more mathematical than graphical methods such as histograms or Q–Q plots. The purpose of the test is to determine whether the fitted distribution is statistically consistent with the data.

The test is based on the null hypothesis
H_0: The data $X_1, X_2, \ldots, X_n$ are IID and follow the fitted distribution $\hat{F}$.
This means that the data come from the fitted distribution.

If the test does not reject H_0, it does not mean that the distribution is perfectly correct. It only means that we did not find strong evidence against it. For small or medium sample sizes, the test is not very sensitive, so small differences may not be detected. For very large sample sizes, the test almost always rejects H_0 because even tiny differences are detected. In simulation studies, we only need a distribution that is close enough, not perfectly exact. Goodness-of-fit tests are mainly used to detect large or gross differences, not small ones.

Chi-Square Goodness-of-Fit Test

The chi-square goodness-of-fit test is the oldest goodness-of-fit test and was proposed by Karl Pearson. It is a formal version of histogram comparison. The main idea is to compare observed frequencies from data with expected frequencies from the fitted distribution. If the observed and expected frequencies are very different, then the fit is considered bad.

Step-by-Step Chi-Square Test

- Step 1: Divide the full data range into k intervals
  $[a_0, a_1), [a_1, a_2), \ldots, [a_{k-1}, a_k)$.
  The intervals may include $(-\infty)$ or $(+\infty)$.

- Step 2: Count the observed data in each interval
  N_j = number of observations in interval j,
  and the total satisfies $\sum$ from j = 1 to k of $N_j = n$.

- Step 3: Compute expected proportions
  Continuous case:
  $p_j = \int$ from a_{j-1} to a_j of $\hat{f}(x)\,dx$
  Discrete case:
  $p_j = \sum$ for values $a_{j-1} \le x_i < a_j$ of $\hat{p}(x_i)$

- Step 4: Compute expected frequencies
  Expected count in interval j = $n p_j$

- Step 5: Compute the chi-square statistic
  $\chi^2 = \sum$ from j = 1 to k of $(N_j - n p_j)^2 / (n p_j)$

Meaning of the Chi-Square Value

If the chi-square value is small, then the observed frequencies are close to the expected frequencies and the fit is considered good. If the chi-square value is large, then there is a big difference between observed and expected values and the fit is considered bad. We reject H_0 if the chi-square value is too large.

Case 1: All Parameters Are Known

This case means that the distribution parameters were not estimated from the data. For example, when testing U(0,1) for random number generators or testing arrival times U(0,T) in a Poisson process, the parameters are known in advance.

If H_0 is true and the sample size is large, then the test statistic follows a chi-square distribution with $(k - 1)$ degrees of freedom. We reject H_0 if
$\chi^2 > \chi^2(k - 1, 1 - \alpha)$,
where α is the significance level such as 0.05, and $\chi^2(k - 1, 1 - \alpha)$ is the critical value from the chi-square table.

The chi-square test is asymptotically valid, which means it works properly only when the sample size n is large.Exam-Friendly Summary

- Goodness-of-fit tests formally check whether data follow the fitted distribution
- Null hypothesis: data follow the fitted distribution
- We do not accept H_0; we only fail to reject it
- Small sample size gives a weak test
- Very large sample size often leads to rejection
- Chi-square test compares observed and expected frequencies
- Large chi-square value indicates a bad fit
- When parameters are known, degrees of freedom = $(k - 1)$

Kolmogorov–Smirnov (K-S) Test

The Kolmogorov–Smirnov test is a goodness-of-fit test. It checks how well your data match a given probability distribution such as normal or exponential. The test compares the empirical distribution function obtained from the sample data with the theoretical distribution function of the assumed distribution.

Earlier, the chi-square test was used for goodness-of-fit testing, but it has several problems. In the chi-square test, the data must be grouped into intervals or bins, which can be confusing when choosing the intervals. Some information is lost during grouping, and the test works well mainly for large samples because it is based on approximation. The K-S test avoids these issues because it does not require grouping and uses the raw data directly.

The basic idea of the K-S test is to compare two curves: the empirical distribution function from the data and the theoretical distribution function of the fitted distribution. The test measures the biggest vertical gap between these two curves. That biggest gap is called the K-S statistic.

The empirical distribution function $F_n(x)$ is formed by sorting the data as $X(1), X(2), \ldots, X(n)$. At each data point, $F_n(X(i)) = i/n$. At the smallest value it equals $1/n$, and at the largest value it equals $n/n = 1$. This creates a step-like curve.

The K-S statistic D_n is defined as the maximum vertical distance between $F_n(x)$ and $\hat{F}(x)$. In simple words, it is the biggest difference between the actual data curve and the theoretical curve. If D_n is small, the fit is good. If D_n is large, the fit is poor.

Problems with Chi-Square Test

- Data must be grouped into intervals (bins)
- Choosing intervals is confusing
- Some information is lost when grouping
- Works well only for large samples

Advantages of K-S Test

- No grouping needed
- Uses raw data directly
- No loss of information
- Works for any sample size n
- Usually more powerful than chi-square
- Very clean and simple idea

Calculation of K-S Statistic

- Sort the data: $X(1), X(2), \ldots, X(n)$
- Compute
  $D_n^1 = \max \mid i/n - \hat{F}(X(i)) \mid$
- Compute
  $D_n^2 = \max \mid \hat{F}(X(i)) - (i - 1)/n \mid$
- Final statistic
  $D_n = \max(D_n^1, D_n^2)$
- Do not use shortcut formulas

Decision Rule

- Reject H_0 if
  $D_n > d_{n,1-\alpha}$
- $d_{n,1-\alpha}$ is the critical value
- α is the level of significance

Case 1: All Parameters Are Known

This case occurs when the distribution parameters are known in advance. For example, testing a Normal(0, 1) distribution where the mean and variance are already known. This is the best case for the K-S test, and one common table works for all continuous distributions. Stephens provided an easy rule: reject H_0 if

$$(\sqrt{n} + 0.12 + 0.11/\sqrt{n})\, D_n > c_{1-\alpha},$$

where the values of $c_{1-\alpha}$ are given in Table 6.15.

Case 2: Parameters Are Estimated from Data

This is the most common case. For example, in a normal distribution where the mean is the sample mean and the variance is the sample variance. In this situation, the original K-S test is not valid. The solution is to use the Lilliefors test with modified critical values. The decision rule becomes

$$(\sqrt{n} - 0.01 + 0.85/\sqrt{n})\, D_n > c'_{1-\alpha},$$

where the values are given in Table 6.15. The same method works for normal, lognormal, exponential, Weibull, and log-logistic distributions.

Limitations of K-S Test

- Not easy to use for discrete data
- Critical values for discrete case are complex
- Original K-S test requires known parameters

The Kolmogorov–Smirnov test compares the empirical distribution function with the theoretical distribution function and uses the maximum vertical distance between them as the test statistic.

6.10 Specifying Multivariate Distributions, Correlations, and Stochastic Processes

So far, we only looked at single random variables, which are called univariate. For example, the service time at a machine can be treated as one random variable, and we usually assume that all random variables are independent. However, in real systems, inputs are often related to each other. This section explains how to handle multiple random variables that are connected and how to deal with processes that change over time.

In many real situations, inputs do not act alone. Multiple variables together may form a multivariate distribution. For example, the service times at two stations for the same part might be correlated. In some cases, we may not know the full joint distribution, but we may know that the variables are correlated. In that case, we still want our simulation inputs to reflect that correlation. Another important case is a stochastic process, which is a sequence of random variables over time. For a stochastic process, we must specify the marginal distribution of each variable and also describe the autocorrelation, which is the relationship between current values and past values.

There are many real-life examples where this is important. In a maintenance shop with two stations such as inspection and repair, if a part is bad, it may require a long inspection and also a long repair time. In this case, the service times are positively correlated. If we ignore this correlation, the simulation results may become inaccurate. In a communication system, messages arrive over time and the sizes of messages may be correlated. Large messages may arrive in groups, creating positive autocorrelation. Ignoring this can produce wrong performance results. In an inventory system, orders over time may be negatively correlated. For example, a large order today may lead to a smaller order tomorrow. These relationships must be considered in simulation.

Ignoring correlations or autocorrelations can make simulation results wrong or unrealistic. A good simulation model requires that inputs reflect the statistical relationships between variables so that the results are valid and meaningful.

When Multivariate Modeling Is Needed

- Random vectors (joint distribution)
- Correlated inputs without full joint distribution
- Stochastic processes over time

Key Takeaways for Exam

- Single random variable → univariate
- Multiple connected variables → multivariate
- Positive correlation → variables move together
- Negative correlation → one increases while the other decreases
- Stochastic process → random variables over time
- Autocorrelation → dependence on past values
- Simulation inputs must reflect known correlations for valid results

When input variables are related or change over time, we must consider multivariate distributions, correlations, or stochastic processes to produce realistic simulation inputs.

6.10.3 Specifying Stochastic Processes

A stochastic process is a sequence of random variables observed over time. Even if each variable has the same marginal distribution, the values may still be correlated with previous values. This time-based dependence is called autocorrelation. In simulation, this is very important because ignoring autocorrelation can produce unrealistic or incorrect results.

For example, consider message sizes arriving at a network node. Each message size may follow the same distribution, but large messages may often be followed by other large messages. This creates positive autocorrelation. If we ignore this relationship and treat message sizes as independent, the simulation results may not correctly represent the real system.

Types of Stochastic Processes Used in Simulation

- AR (Autoregressive) processes

- Gamma and EAR (Exponential Autoregressive) processes
- ARTA (Autoregressive-To-Anything) processes
- VARTA (Vector ARTA) processes

AR (Autoregressive) Process

An AR(p) process means each value depends on the previous p values plus a random noise term. The random noise is usually normally distributed. This process works best when the marginal distribution is normal. It is often used as a basic building block for more flexible models. The idea is simple: the present value is influenced by past values.

Gamma and EAR Processes

Gamma processes have a Gamma marginal distribution and include autocorrelation. EAR processes have an Exponential marginal distribution with autocorrelation. Both are constructed in a way similar to AR processes but are designed for specific marginal distributions. They are useful when the data clearly follow Gamma or Exponential distributions.

ARTA (Autoregressive-To-Anything) Process

ARTA is one of the most flexible models for simulation input. It can model any stationary marginal distribution along with a specified autocorrelation structure.

The idea is to first generate a normal AR process. Then this normal process is transformed into any desired distribution using the inverse transform method. First, the normal values are converted into uniform values between 0 and 1. Then those uniform values are transformed into the required distribution. This method allows ARTA to match both the marginal distribution and the autocorrelation up to a chosen lag.

VARTA (Vector ARTA) Process

VARTA is an extension of ARTA for multivariate stochastic processes. Instead of generating one variable over time, it generates a vector of correlated variables at each time point. It allows us to specify marginal distributions for each variable and also specify autocorrelations within each variable and correlations across different variables. It is based on a Gaussian vector autoregressive base process and is used when simulation inputs are both time-dependent and cross-correlated.

Key Exam Points

- Stochastic process → sequence of correlated random variables over time
- Autocorrelation → dependence on past values
- AR process → normal marginal distribution, depends on previous p values
- Gamma / EAR → specific marginal distributions with autocorrelation

- ARTA → flexible, matches any marginal distribution and autocorrelation
- VARTA → multivariate extension handling correlated vectors over time

To simulate correlated inputs over time, AR, Gamma/EAR, ARTA, or VARTA processes are used depending on whether the input is univariate or multivariate and on the required marginal distributions and autocorrelation structure.

What is an arrival process?

- In many simulations, we track **events happening over time**, e.g., customers arriving.
- Let (t_i) = time of the (i)-th arrival.
- Interarrival time: ($A_i = t_i - t_{i-1}$) → time between consecutive arrivals.
- ($N(t)$) = number of arrivals **up to time t**.

Basically: An **arrival process** is a stochastic process describing **how events (customers, messages, etc.) occur over time**.

Poisson Process (Sec 6.12.1)
A Poisson process is an arrival process where:

1. Customers arrive **one at a time**.
2. Number of arrivals in a time interval **doesn't depend on previous arrivals** (independence).
3. The **arrival rate is constant** over time.

Key properties:

- Number of arrivals in an interval of length (s) → **Poisson distributed** with mean (λs).
- Interarrival times ($A_1, A_2, \ldots$) → **IID exponential** with mean ($1/\lambda$).
- (λ) = expected number of arrivals per unit time.

Important: If interarrival times are IID exponential → arrivals form a Poisson process (converse also true).

Example: Cars arriving at a drive-up bank during a 90-min period → approx. exponential interarrival times → Poisson arrivals.

Nonstationary Poisson Process (Sec 6.12.2)

Why nonstationary?

- Real systems often have **varying arrival rates**.

- Example: Fast-food restaurant → high at noon, low in afternoon.
- Freeway traffic → peaks in morning & evening.
  A nonstationary Poisson process:

1. Customers arrive **one at a time**.
2. Numbers of arrivals in **disjoint intervals are independent**, but the **rate (\lambda(t)) changes over time**.

- Let (L(t) = E[N(t)]) = expected number of arrivals by time t.
- Then (\lambda(t) = dL(t)/dt) → instantaneous rate at time t.
- Number of arrivals in ([t, t+s]) is **Poisson** with mean (b(t,s) = \int_t^{t+s} \lambda(y) dy).

Example: Copy shop arrivals (11 a.m. – 1 p.m.)

- Divide 2 hours into 12 subintervals (10 min each).
- Average arrivals per subinterval over 8 days → estimate of (\lambda(t)).
- Plot of (\lambda(t)) shows **how arrival rate changes over time**.

Subinterval length matters: too small → noisy estimates; too large → lose details.

How to estimate (\lambda(t)) / L(t)

1. **Piecewise-constant method** → divide time into intervals, assume constant rate in each.
2. **Piecewise-linear / polynomial** → smoother, allows trends and cycles.
3. **Parametric functional form** → assume a mathematical model for (\lambda(t)) and estimate parameters (max likelihood, least squares).
4. **Combined nonparametric/parametric** → flexible for long-run trends, cycles, nontrigonometric effects.

These methods let us generate realistic arrival times in simulations.Key exam points

1. **Arrival process** = events over time, interarrival times = (A_i).
2. **Poisson process**: IID exponential interarrival times, constant rate.
3. **Nonstationary Poisson**: rate (\lambda(t)) varies with time; still independent arrivals.
4. Estimation of (\lambda(t)) → piecewise constant, linear, or parametric models.
5. Realistic modeling of arrivals is crucial → affects simulation accuracy.

A Poisson process models random arrivals with exponential interarrival times; if the rate changes with time, use a nonstationary Poisson process with a time-dependent rate (\lambda(t))..